

Raj Reddy

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Education

University of Utah

B.S. Computer Science

Salt Lake City, UT

August 2020 – May 2025

Relevant Coursework: Computer Systems, Machine Learning, Computer Graphics, Algorithms, Software Practice I & II, Database Systems, Computer Networks, Foundations of Data Analysis, Models of Computation, and Linear Algebra.

Skills

Languages: Python, TypeScript, C++, SQL, Java

Software: Pandas, NumPy, React, Node.js, AWS, Azure, Docker, Next.js, .NET, REST, Django, MongoDB, Linux

Experience

L3Harris Technologies

Associate Software Engineer

Dallas, TX

June 2025 – Present

- Built containerized microservices for high-frequency sensor data ingestion and processing using Podman and serverless infrastructure on AWS, designing storage schemas and retrieval patterns optimized for high-throughput distributed workloads.
- Developed Python and C++ signal processing pipelines on embedded Linux systems, applying distributed data flow patterns to route, transform, and persist live sensor streams under strict latency and reliability constraints.
- Automated CI/CD infrastructure with static analysis and integration testing, maintaining deployment reliability across multiple production services using infrastructure-as-code tooling.

Doxy.me

Software Engineer

Charleston, SC

August 2024 – May 2025

- Designed and operated distributed ML inference pipelines on AWS SageMaker and Fargate, building event-driven data flows that ingested WebRTC video streams, processed them through TypeScript and Python services, and persisted structured outputs for downstream analytics.
- Owned cloud infrastructure configuration and deployment automation using YAML-based pipelines, managing containerized services across multiple production environments with zero-downtime release strategies.
- Built observability into production pipelines, monitoring latency and throughput across distributed services and iterating on model accuracy using reinforcement learning feedback loops.

University of Utah

Undergraduate Research Assistant – FuTURES Lab

Salt Lake City, UT

May 2024 – Jun 2025

- Analyzed and optimized large-scale datasets, enhancing software configuration testing with gcov and CMake, increasing code coverage by 30% on real-world APIs (Libpng, PyTorch).
- Expanded OSS-Fuzz testing coverage using compile-time options to improve the robustness of full-stack libraries.

Projects

University Course Planning/Review Platform (Class Best): Built a multi-service data platform enabling student schedule matching, course overlap detection, and behavioral insights. Developed ETL pipelines in Python to process thousands of course records, user preferences, and time-series activity logs. Built a responsive frontend in React + Tailwind with interactive dashboards, heatmaps, and calendar visualizations.

Configuration Fuzzer: Developed a configuration fuzzing tool for OSS-Fuzz, automating build generation to identify critical compile-time configurations and improve code coverage, uncovering previously untested code paths and increasing bug detection effectiveness.

Telehealth Mental Health Diagnosis Tool (Doxy.me): Built an AI system that analyzes visual, speech, and tonal cues from telehealth sessions using PyTorch and Llama, delivering structured mental health insights to treating physicians via SageMaker and Fargate inference pipelines. Improved diagnostic accuracy using reinforcement learning and WebRTC video integration.